

Program : Diploma in Polymer Technology	
Course Code : 3096	Course Title: Polymer Science Lab
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To provide knowledge and skill in standardization and estimation of solutions.
- To provide ability to do polymerization process and determination of specific gravities of samples.

Course Prerequisites:

Topic	Course code	Course name	Semester
Normality calculation		Mathematics I & II	1 & 2
Titrations and standardization		Applied Chemistry	1

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Determine the normality and weight per litre of solute of solutions by titration methods	19	Applying
CO2	Produce polymers by different polymerization techniques	15	Applying
CO3	Determine the specific gravity of plastics and elastomers	9	Applying
	Lab Exam	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3			3			
CO3		3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	To determine the normality and weight per liter of solute of solutions by titration methods.		
M1.01	Familiarization and study of analytical balance	3	Applying
M1.02	Determine the normality of given hydrochloric acid using a standard solution of sodium carbonate.	3	Applying
M1.03	Determine the normality of given NaOH using a standard solution of H ₂ C ₂ O ₄	3	Applying
M1.04	Estimate the amount of sodium hydroxide, provided with AR sodium carbonate using a link solution of hydrochloric acid.	5	Applying
M1.05	Estimate the amount of HCl, provided with AR oxalic acid using a link solution of sodium hydroxide	5	Applying
	Lab Exam I	1	
CO2	To produce polymers by different polymerization techniques		
M2.01	Carry out the preparation of UF resin	3	Applying
M2.02	Carry out the preparation of PF resin	3	Applying
M2.03	Carry out the preparation of PMMA by bulk polymerization	3	Applying
M2.04	Carry out the preparation of Regenerated cellulose	3	Applying
M2.05	Carry out the preparation of polystyrene by emulsion polymerization	3	Applying

CO3	To determine the specific gravity of plastics and elastomers		
M3.01	Find out the specific gravity of polymer sample by Hydrostatic bench method >1	3	Applying
M3.02	Find out the specific gravity of polymer sample by Hydrostatic bench method, <1	3	Applying
M3.03	Find out the specific gravity of polymer sample by bottle method	3	Applying
	Lab exam II	1	

Text / Reference:

T/R	Book Title/Author
1	1. Polymer science -V.R.Gowarikar (Publisher – New Age International (P) Ltd, New Delhi- 2005 Edition.
2	Prof A.O.Thomas –Practical chemistry-Eight edition 2000
3	ASTM standards

Web Source Reference:

Sl.No	Website Link
1	https://sciencing.com
2	http://www.sciencedirect.com
3	http://engineerfeed.com
4	https://en.m.wikipedia.org